

Annex 4: CAN Filter Settings

Instruction Manual

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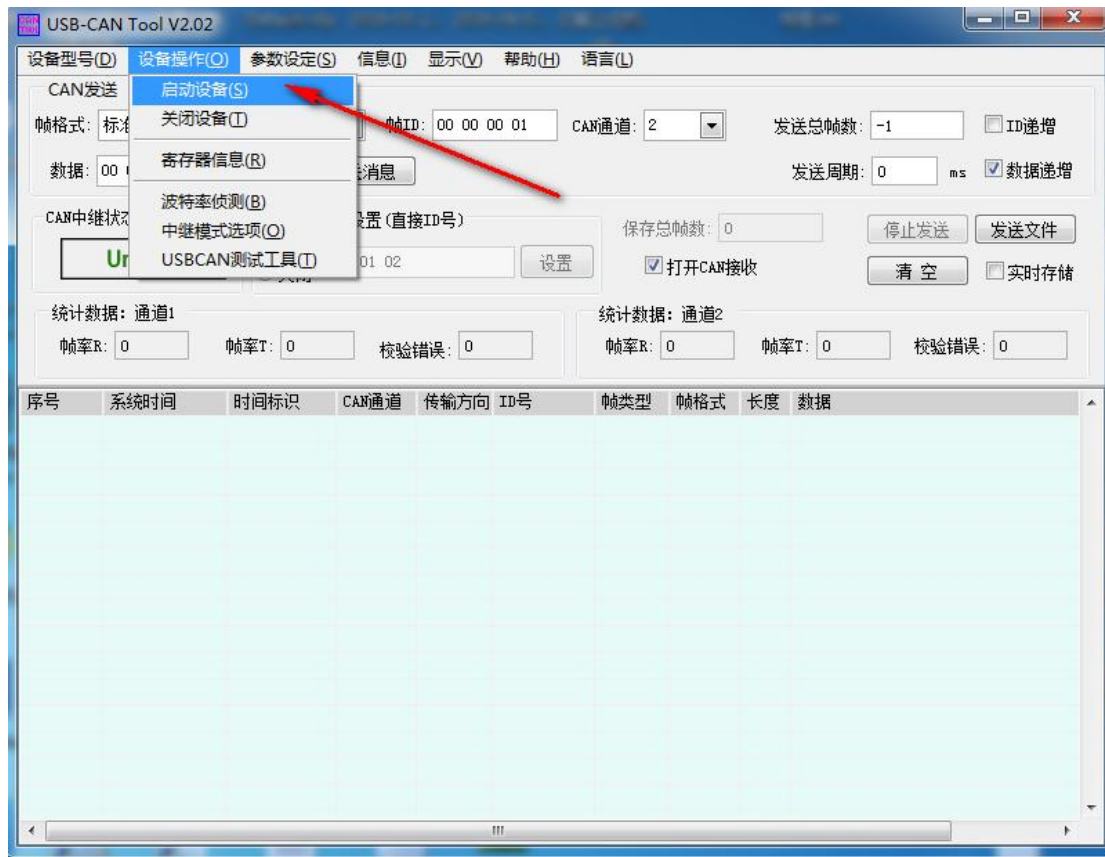
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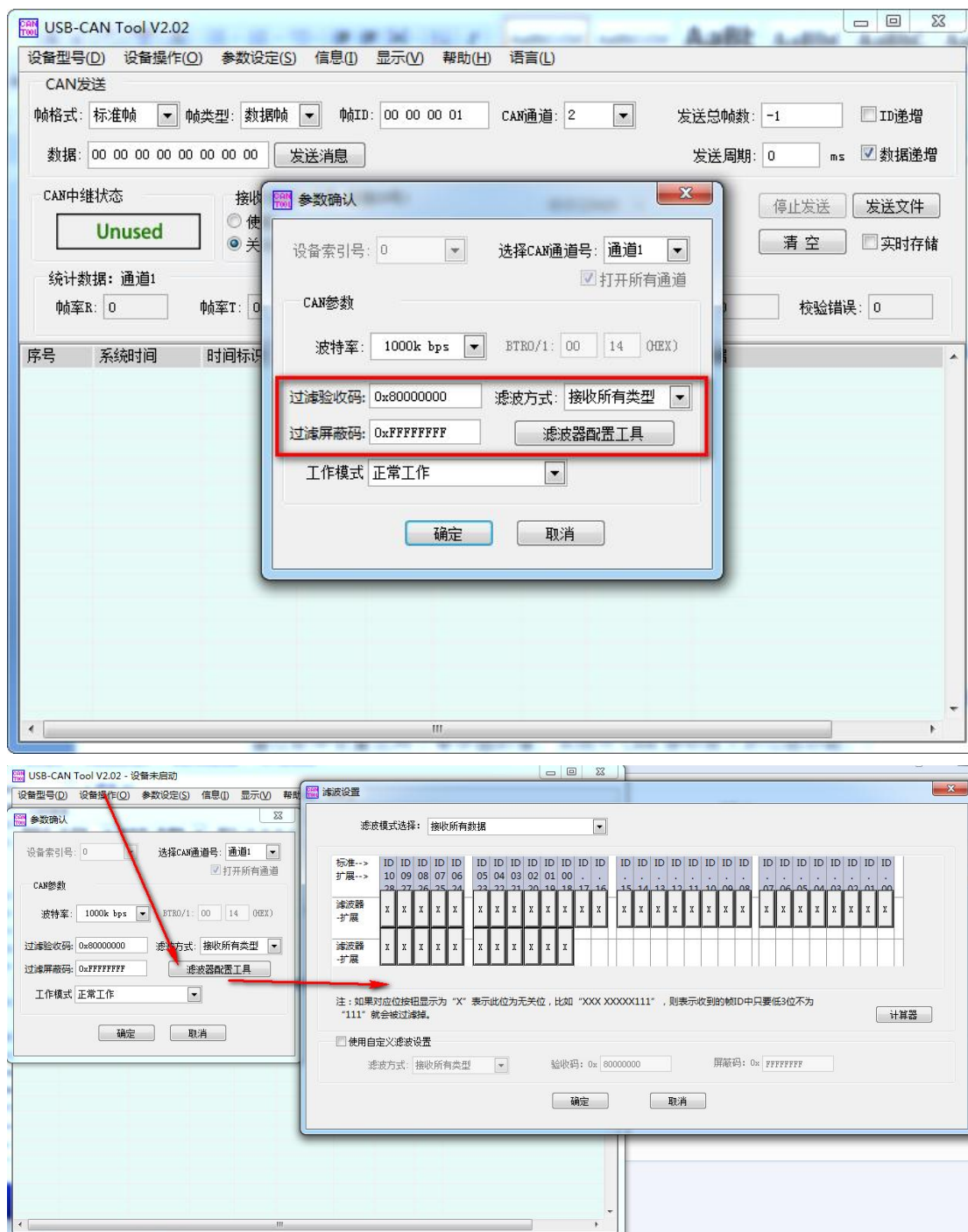
I. Filter Configuration

1 The configuration of the Filter in Each Debugging Software

1.1 Configuration of the Original Debugging Tool USB_CAN TOOL Software

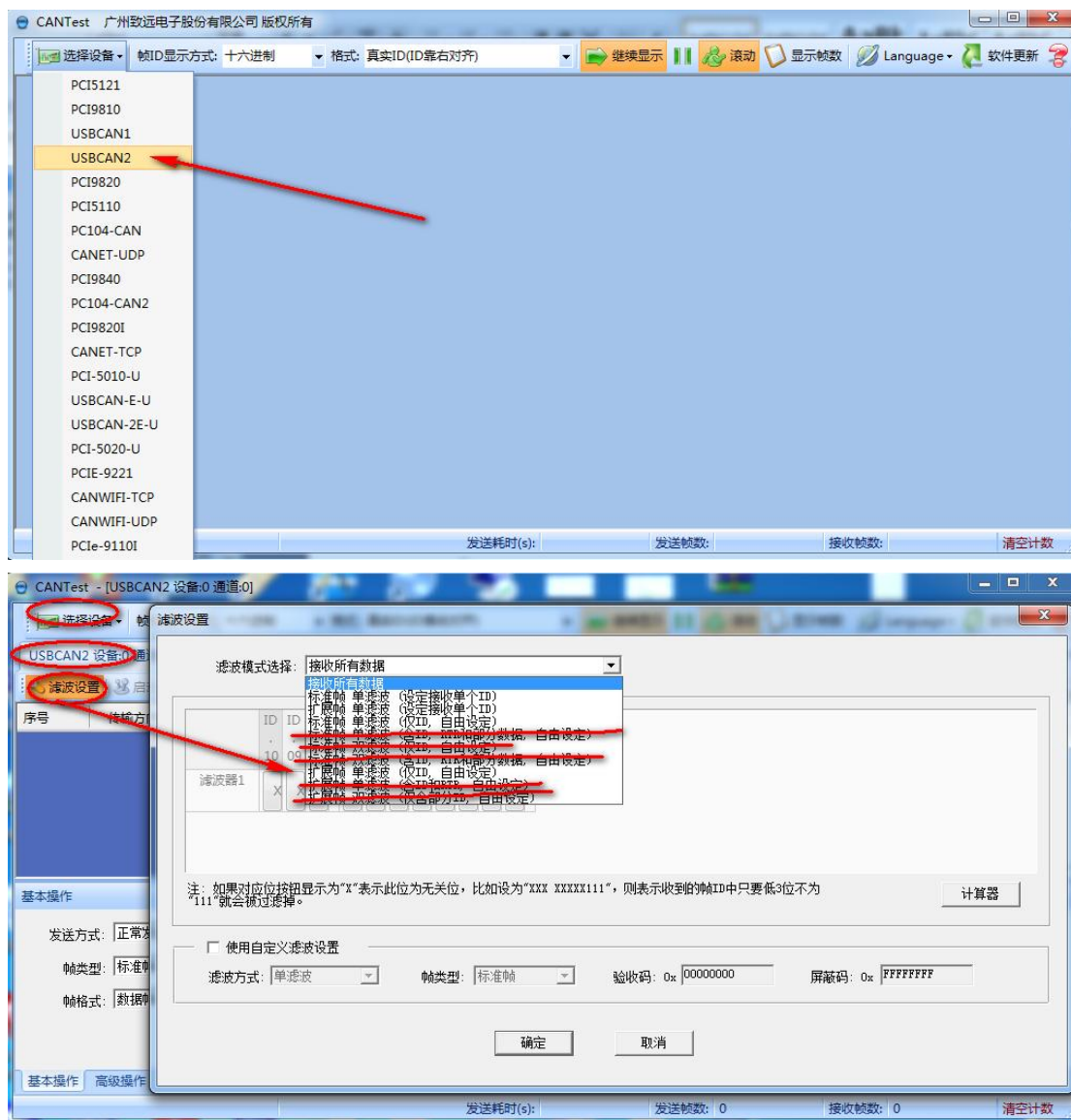
Device Operation->Start the device, in the pop-up parameter configuration interface, there are three parameters: "Filter Acceptance Code", "Filter Masking Code", "Filter Method". These three parameters can be configured intuitively through the "Filter Configuration Tool".





1.2 Chou-Li-Gong Debugging Software CANtest Software (Model USBCAN, USBCAN2)

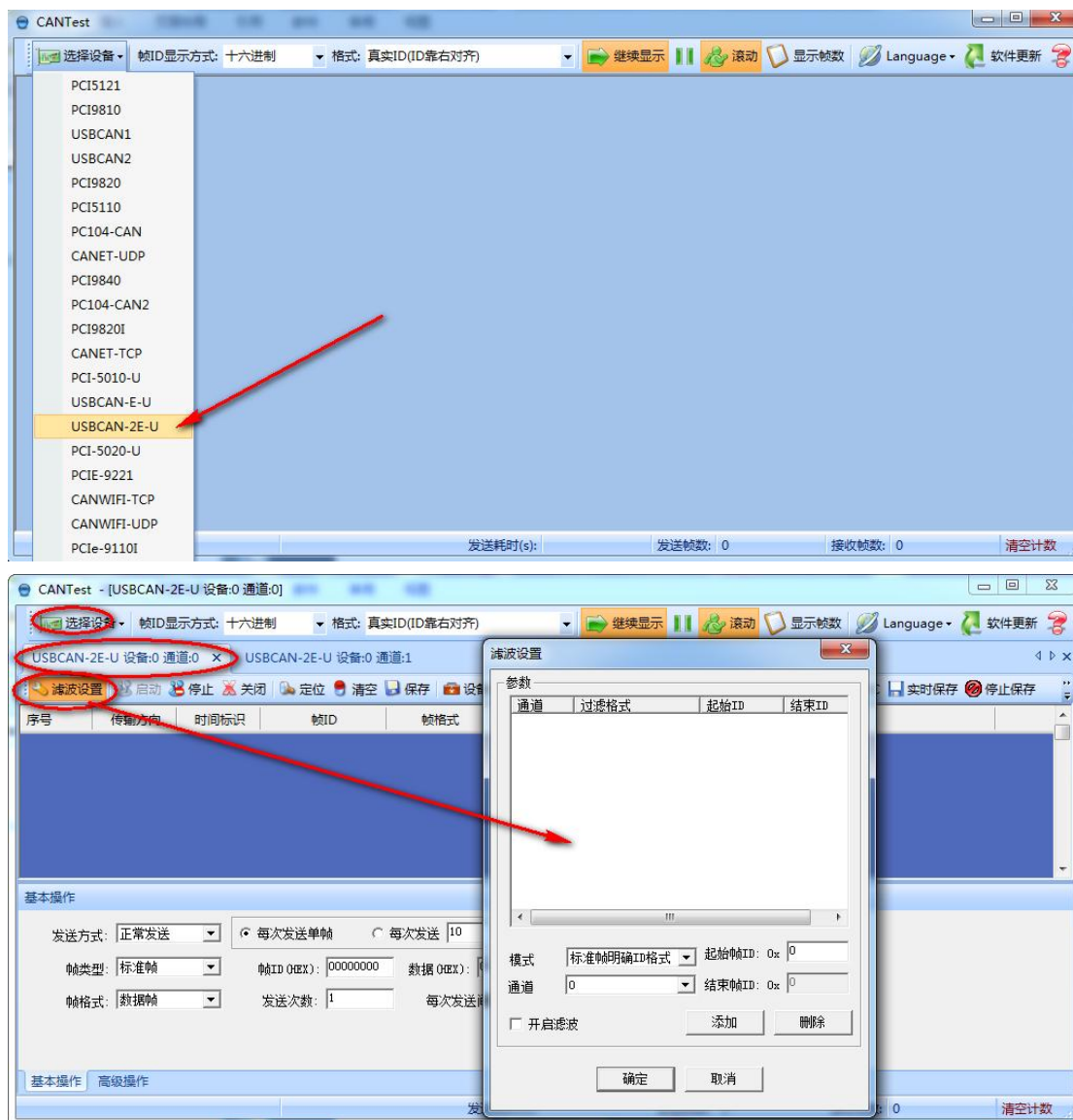
Select Device->USBCAN, USBCAN2, configure baud rate and other parameters in the pop-up parameter configuration interface, and then start the device. After starting the device, each channel has a corresponding filter setting item. There are four parameters: "Filter mode", "Frame type", "Acceptance code" and "Mask code". These four parameters can be configured through the "Filter Method", "Frame Type", "Acceptance Code" and "Masking Code". These four parameters can be configured intuitively through the "Filter Setting Interface".



Note: The Parts Crossed out in Red Are Not Optional.

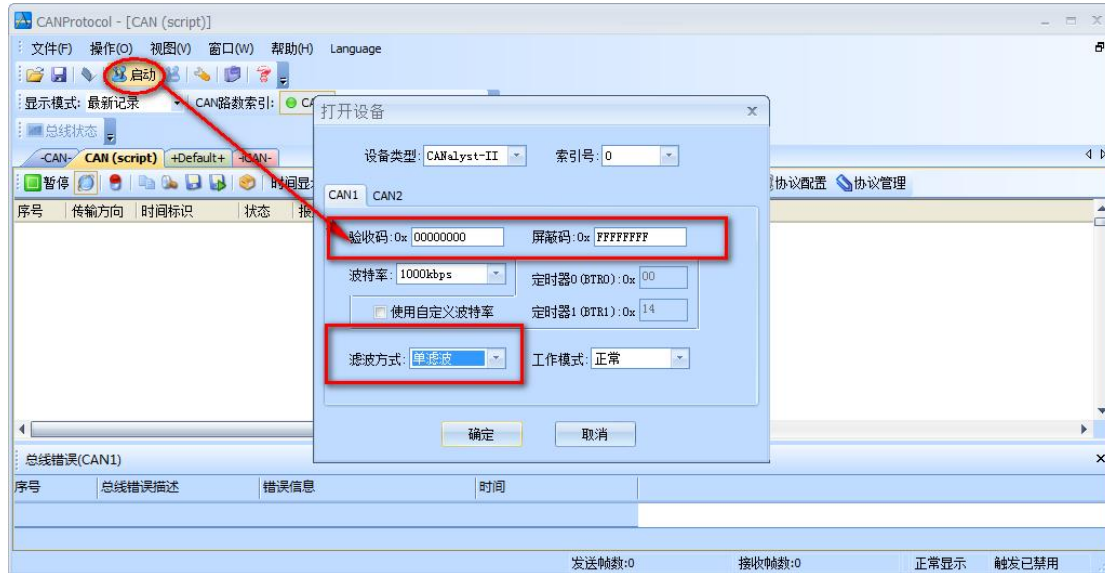
1.3 Chou-Li-Gong Commissioning Software CANtest Software (Model USBCAN-E-U, USBCAN-2E-U)

Select Device->USBCAN-E-U, USBCAN-2E-U, and start the device after configuring baud rate and other parameters in the pop-up parameter configuration interface. After starting the device, there are corresponding filter setting items for each channel (all are the same plug-in, and you can configure the filters of two channels at the same time under one channel interface). There are four parameters: "Mode", "Channel", "Start ID" and "End ID". These four parameters can be configured intuitively through the "Filter Setting Interface".



1.4 Chou-Li-Gong Commissioning Software CANPro Software

Operation->Start the system, in the pop-up parameter configuration interface, there are three parameters: "Filter Acceptance Code", "Filter Masking Code", "Filter Mode" three parameters. You can only choose single filter for "Filter mode".



To summarize: The configuration of the original debugging tool USB_CAN TOOL software has the best compatibility. There is a compatibility problem between the software CANtest (model USBCAN, USBCAN2) and the software CANPro. The filter configuration of the Chou-Li-Kong CANtest software (model USBCAN-E-U, USBCAN-2E-U) is the most intuitive and convenient, and is recommended.

2 Configuration of Filtering in the Secondary Development Library

2.1 Secondary Development (Model USBCAN, USBCAN2)

When setting up the filter acceptance filter, use the AccCode parameter of the VCI_InitCan function to pass; when setting up the filter mask register, use the AccMask parameter of the VCI_InitCan function to pass; when setting up the filtering mode, use the Filter parameter of the VCI_InitCan function to pass; for specific applications, please refer to "Interface Function Library (Secondary Development Library) Specific applications can refer to *"Interface Function Library (Secondary Development Library).pdf"*.

2.2 Secondary Development (Model USBCAN-E-U, USBCAN-2E-U)

VCI_SetReference

VCI_SetReference(DWORD DevType, DWORD DevIndex, DWORD CANIndex, DWORD RefType, PVOID pData);

II. Filter Configuration Details

1 Filter Registers

The filter register consists of two 32-bit registers: the filter acceptance filter (ACR) and the filter mask register (AMR). The filtering function for CAN received messages is realized by setting the values of these two registers through software.

When the CAN module receives a message, it compares the message identifier with the corresponding bit in the filter. If the identifier matches the user-configured filter, the message is stored in the corresponding receive cache queue of the CAN controller.

The receive mask can be used to ignore the select bits of the identifier during reception. These bits will not be compared to the bits in the filter when the message is received. For example, if the user wishes to receive all messages with identifiers 0, 1, 2, and 3, the user would need to mask out the lower 2 bits of the identifier. A bit of the Masking Register equal to 1 indicates that the filtering of the bit corresponding to the ID bit is ignored, e.g., if the Masking Register value = FFFFFFFF, then all messages can be received.

The acceptance filter ACR, and the acceptance mask AMR are both 32bits (4bytes). For ID values that require acceptance filtering, the highest bit of the ID (Bit10 for the highest bit of the ID in standard frames, Bit28 for extended frames) is aligned with the highest bit (Bit31) of the ACR/AMR, i.e., **left-aligned**.

Both the CAN bus acceptance filter and the acceptance mask are for CAN reception. Note: When AMR is FF FF FF FF FF, it means that all filter bits of ACR are masked, i.e. all messages can be received.

Note: For a detailed description of the ID format, please refer to: "*Annex 1: ID Alignment.pdf*" instruction document.

Examples:

(1) . Standard \Frame Example: To receive standard frames, the filtering method needs to be selected as "Receive all types" or "Receive standard frames only", ACR=Any value, AMR= 0xFFFFFFFF, the adapter can receive CAN messages with

2 Filter Modes

The USB-CAN adapter provides three filter modes: receive all types, receive only standard frames, and receive only extended frames. The following table shows the comparison of filter mode values:

Value	Name	Clarification
0/1	Receive All Types	The filter filters both standard and extended frames!
2	Standard Frame Only	The filter only filters standard frames, extended frames will be filtered out directly.
3	Extended Frame Only	The filter only filters extended frames, standard frames will be filtered out directly.

In setting the filter mode, VCI_SetReference2 function of the RefType parameter must be passed 1, the specific application can refer to the sample code after this subsection.

2.1 Receive All Types

When the filter mode is set to "Receive All Types", the filter acceptance register and mask register settings are valid for both standard frame ID and extended frame ID, and the mapping method is left-aligned.

For example, in this filtering mode, the value of acceptance code = 0xF8000000, the value of mask code = 0x01FFFFFF, then according to the left-aligned mapping, when receiving an extended frame, the low 22 bits of its ID will be ignored, and the high 7 bits must match with the high 7 bits of the acceptance code in order to be received; similarly, when receiving a standard frame, the low 4 bits of the ID will be ignored, and the high 7 bits must match with the high 7 bits of the acceptance code in order to be received. must match the acceptance code high 7 bits to be received. As shown in the figure below:



2.2 Receiving Standard Frames Only

When the filter mode is set to "receive standard frame only", the settings of the filter acceptance register and mask register are valid for the standard frame ID, and the received extended frame is directly ignored, and the mapping method is left-aligned.

For example, in this filtering mode, the value of acceptance code = 0x3F000000, the value of mask code = 0x00FFFFFF, then according to the left-aligned mapping, when receiving a standard frame, the lower 3 bits of its ID will be ignored, and the higher 8 bits must match with the higher 8 bits of the acceptance code before it is received; **for any extended frames received under the filtering mode, they will be directly blocked by the filter.** As shown in the figure below:

滤波模式选择: 标准帧滤波 (仅ID, 自由设定)

ID	ID	ID	ID	ID	ID	ID	ID	ID	ID	ID
10	09	08	07	06	05	04	03	02	01	00
0	0	1	1	1	1	1	1	X	X	X

注: 因标准帧ID只有11位, 所以验收码和屏蔽码的32位寄存器, 在“只接收标准帧”滤波方式下, 只有高11位有效, 其余位无论设置何值, 均不影响滤波效果!

标准帧ID低3位不参与滤波

注: 如果对对应位按钮显示为“X”表示此位为无关位, 比如“XXX XXXXX111”, 则表示收到的帧ID中只要低3位不为“111”就会被过滤掉。

使用自定义滤波设置

滤波方式: 只接收标准帧

验收码: 0x 3F000000

屏蔽码: 0x 00FFFFFF

只接收标准帧

屏蔽码高8位为0, 则过滤标准帧ID高8位

确定 取消

2.3 Receive Extended Frames Only

When the filter mode is set to "Extended frame only", the settings of the filter acceptance register and mask register are valid for extended frame IDs, while the received standard frames are directly ignored, and the mapping method is left-aligned.

For example, in this filtering mode, the value of acceptance code = 0xFFFF40000, the value of mask code = 0x0003FFFF, then according to the left-aligned mapping, when receiving an extended frame, the lower 15 bits of its ID will be ignored, and the higher 14 bits must match with the higher 14 bits of the acceptance code in order to be received; for any standard frames received in this filtering mode, they will be blocked directly by the filter. As shown in the figure below:



Caution:

When the CAN bus is sending and receiving data normally, try not to modify the CAN bus parameters or turn off the CAN bus through the USBCAN adapter, and wait until the data sending and receiving stops or disconnect the USBCAN adapter from the CAN bus before carrying out the corresponding operation.